staunton

Chess diagrams, games and tournament tables for Typst

`https://github.com/ndg6/staunton`

User manual · package version 0.1.0

Guide

API Reference

1 Introduction

Package **staunton** aims to provide a complete, convenient but also flexible solution for chess publications. It provides a full set of features, including:

- **boards and diagrams** — pure boards with labels, highlights, arrows, an optional grid, flexible sizing, custom colors, and bundled SVG piece sets (or a Unicode fallback); and building on that diagrams with captions, figure counters, and referenceable labels;
- **games from PGN** — a sophisticated parser creates single games or an array of games, from which you create positions by "locators" (mainline and variations), move play-out, and FEN export;
- **move notation** — from parsed games you get move text output with localized piece letters, figurine glyphs, NAGs, comments and diagrams embedded inline;
- **tournament tables** — we can create standings, cross-tables and progress charts from a PGN's results, by player or by team;
- **outlines and references** — diagrams and tables get their own counters and lists;
- **document-wide styling** and **localization** (six languages, easily extended);
- **limited HTML export** — notation, tables, outlines, references and captioned figures become native HTML, with boards and diagrams embedded as inline SVG (see Section 9).

```
#chess-diagram(
  "r1bqkbnr/pppp1ppp/2n5/4p3/4P3/5N2/
PPPP1PPP/RNBQKB1R",
  caption: [The Ruy Lopez, three moves
in.],
  size: 4cm,
)
```

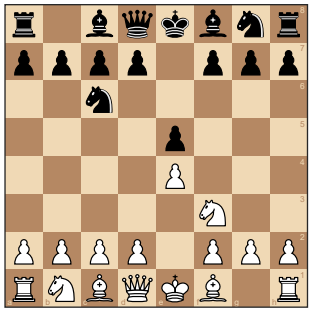


Diagram 1: The Ruy Lopez, three moves in.

This manual is bottom-up. The **board** is the drawing primitive; a **diagram** wraps a board in a referenceable figure; a **position** is the data a board draws; **games** add PGN, notation, and play; then come **tournament tables**, **outlines and references**, and the **document-wide defaults**.

1.1 Installing and Importing

staunton is a Typst package. Import its public API once and every function in this manual is in scope:

```
#import "@preview/staunton:0.1.0": *
```

1.2 How to read this manual

In every framed example, the left side is **the code you type** and the right side is **exactly what it renders** — the manual compiles its own examples, so the two can never disagree.

1.3 The Name

Typst package **staunton** is named in honour of **Howard Staunton** (c. 1810–1874): a leading chess master of his day, organiser of the first international tournament (London, 1851), a chess author and publisher, and the namesake of the standardised **Staunton pattern** chessmen — still the tournament standard.

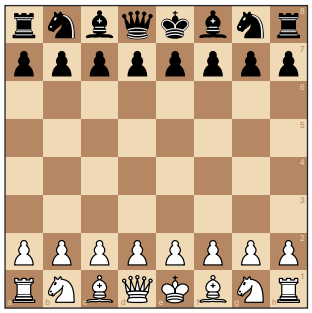
1.4 Acknowledgements

The Typst package boards-n-pieces was an inspiration for some of staunton's features. This package and its manual were developed with assistance from Claude (Opus 4.8) by Anthropic.

2 The Board

`board(source, ..)` draws **just the board** — no caption, no figure — and is the primitive every diagram builds on. `source` is one of: a **FEN string**; a **position** (from `position(..)` or `parse-fen(..)`); or a bare **squares** dict (square name ⟶ `(kind, color)`).

```
#board(
  "rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/
RNBQKBNR",
  size: 4cm,
)
```



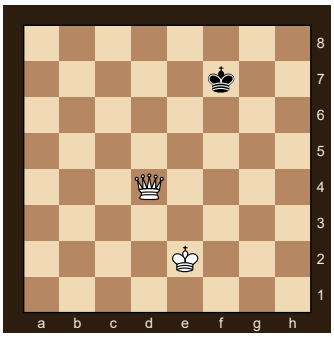
`board` is the variant-agnostic primitive; `chess-board` is the standard-chess sugar over it (same rendering, but it documents the variant and rejects a non-standard source). Use `board` inline in text or inside your own layout; reach for a **diagram** (next chapter) when you want a captioned, referenceable figure.

The rest of this chapter covers the board's drawing options: labels, highlights, arrows, the grid, coordinates, size, colors, orientation, and piece sets — all of which a `chess-diagram` accepts too.

2.1 Labels

`label-mode` chooses how labels for files and ranks are drawn — `"on-square"` (default, tucked into the corner of the squares), `"outside"` (a gutter strip), or `"border"` (a themed band, styled by `border-theme`). `labels: false` suppresses labels completely.

```
#board(
  "8/5k2/8/8/3Q4/8/4K3/8",
  label-mode: "border",
  border-theme: "brown",
  size: 3.8cm,
)
```



In `"border"` mode, `border-theme` picks the band's look — the fill color and the contrasting label color:

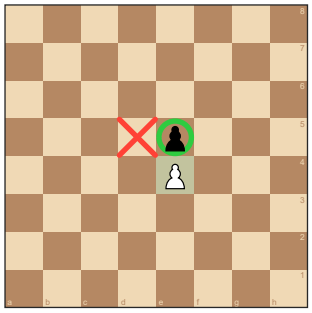
- `"square"` (default) — the band reuses the board's own `dark` square color with `light`-colored labels, so the border blends into the board;
- `"brown"` — a very dark-brown band with creme labels (a warm, classic frame);
- `"dark"` — a charcoal band with light-grey labels (suits dark backgrounds).

`border-theme` is a normal board option: set it per call as above, or document-wide with `set-board-defaults(border-theme: ..)` / `set-chess-defaults` (see Section 8). It only takes effect with `label-mode: "border"`.

2.2 Highlights

`highlight` marks squares; each entry is a square name (drawn with `highlight-shape`, default `"filled"`), a `(square, color)` pair, or a dict `(square: .., shape: .., color: ..)` where `shape` is `"filled"`, `"cross"`, or `"circle"`. By convention a **cross** marks an empty square.

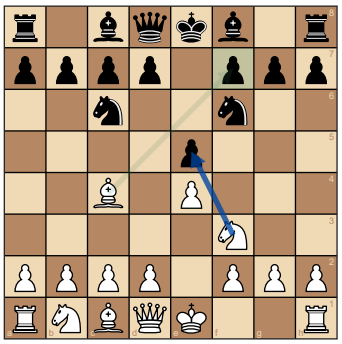
```
#board(
  "8/8/8/4p3/4P3/8/8/8",
  highlight: (
    "e4",
    (square: "e5", shape: "circle"),
    (square: "d5", shape: "cross"),
  ),
  size: 4cm,
)
```



2.3 Arrows and the Grid

As the name suggests `arrows` draws arrows on the board; each entry is a `(from, to)` or `(from, to, color)` tuple, or a dict `(from: .., to: .., color: ..)`. A missing color uses `arrow-color`. Arrows scale with the board and flip with it. A `grid: true` overlay draws thin lines between the squares.

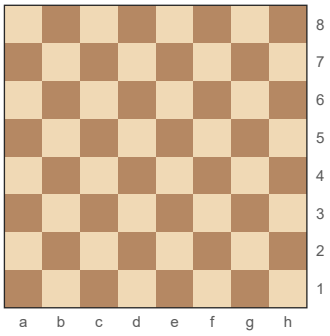
```
#board(
  "r1bqkb1r/pppp1ppp/2n2n2/4p3/2B1P3/5N2/PPPP1PPP/RNBQK2R",
  grid: true,
  highlight: ("f7",),
  arrows: (("c4", "f7"), ("f3", "e5", rgb(0, 70, 160, 200))),
  size: 4.4cm,
)
```



2.4 Coordinates and Non-Square Boards

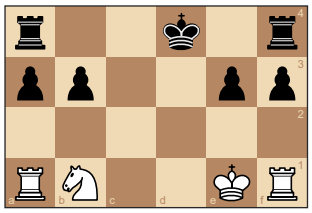
At leasst in standard western chess, Files run `a`, `b`, ... and ranks `1`, `2`, ...; `a1` is the dark square in the lower-left corner, `h8` the upper-right. Square names are case-insensitive (`"E4"` = `"e4"`).

```
#board(
  "8/8/8/8/8/8/8/8",
  label-mode: "outside",
  size: 4cm,
)
```



But boards are **not** tied to 8×8. A `position` built from the string form (next chapter) counts its own columns and rows, and the renderer draws whatever geometry it is given — files and ranks extend as far as the board needs, and the cells stay square while the board itself becomes rectangular:

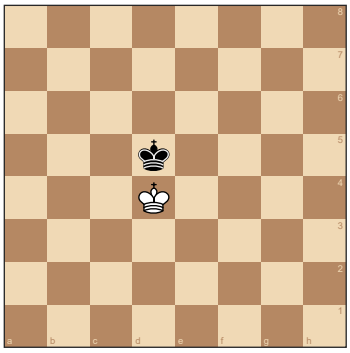
```
#board(
  position(
    "r..k.r",
    "pp..pp",
    "......",
    "RN..KR",
  ),
  size: 4cm,
)
```



2.5 Sizing

The default board size suits an A4 two-column layout. A board reads the available space at its insertion point via `layout`, so it adapts to any column or page size without being told the geometry, and shrinks to fit if asked for more than fits. `size` may be a `length`, a `ratio` of the available width, or `auto`:

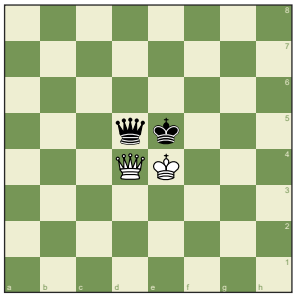
```
#board(
  "8/8/8/3k4/3K4/8/8/8",
  size: 60%,   // of the available
width
)
```



2.6 Colours

`light` and `dark` set the two square colors:

```
#board(
  "8/8/8/3qk3/3QK3/8/8/8",
  light: rgb("#eeeed2"),
  dark: rgb("#769656"),
  size: 3.8cm,
)
```

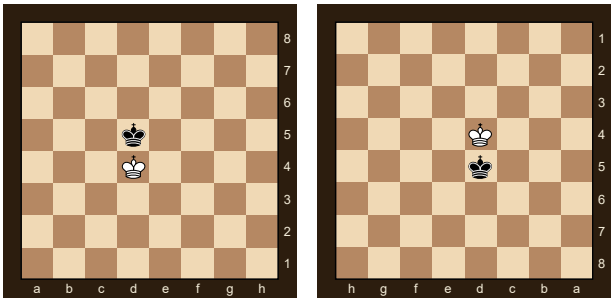


2.7 Flip

`flip: true` shows the board from Black's side; labels, highlights and arrows all flip with it. Orientation is a **per-board** choice — `flip` is the one setting that cannot be made a document default (see **Document-wide style**).

Shown side by side with `"border"` labels, the flipped coordinates are easy to spot — `a1` moves from the lower-left to the upper-right:

```
#grid(columns: 2, gutter: 8pt,
  board("8/8/8/3k4/3K4/8/8/8",
    label-mode: "border", border-theme: "brown", size: 3.4cm),
  board("8/8/8/3k4/3K4/8/8/8", flip: true,
    label-mode: "border", border-theme: "brown", size: 3.4cm),
)
```

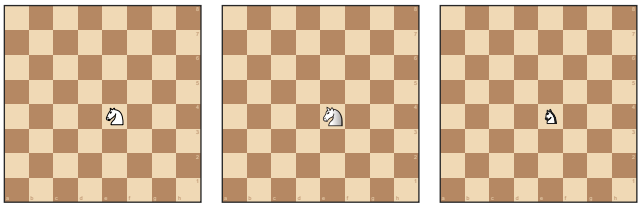


2.8 Piece Sets and Fonts

Pieces are normally drawn from bundled **SVG piece sets** — the preferred rendering: vector art that stays crisp at any board size and looks the same across platforms. A **Unicode glyph** set is provided only as a **fallback** (see below), for when you want no SVG dependency or a font-native look.

Two SVG sets ship with the package: `cburnett` (default) and `merida`. Both are distributed under a license that **permits commercial use** (GPLv2+) — a deliberate constraint: staunton only bundles piece art whose license does not forbid commercial publication, so you can use it freely in commercial work. Choose one with `piece-set:` per board, or document-wide with `set-piece-set` (see Section 8):

```
#grid(columns: 3, gutter: 8pt, align: bottom,
  board("8/8/8/8/4N3/8/8/8", size: 2.6cm, piece-set: "cburnett"),
  board("8/8/8/8/4N3/8/8/8", size: 2.6cm, piece-set: "merida"),
  board("8/8/8/8/4N3/8/8/8", size: 2.6cm, piece-set: "unicode"),
)
```



The renderer accepts **any** set name and loads a piece's SVG on demand from

`src/assets/piece_sets/<name>/{w,b}{K,Q,R,B,N,P}.svg`

so you can add a set by dropping such a folder into your copy and passing its name. `piece-set: "unicode"` (or `none`) selects the glyph fallback — solid Unicode chess glyphs distinguished by fill and a contrasting stroke; it needs a font carrying them.

The rank/file **labels** are drawn in their own sans-serif, set by the `label-font` board option (a family or a fallback list), independent of the document font. The default is `("Arial", "DejaVu Sans Mono")` — Arial on Windows/macOS, falling back to Typst's always-embedded mono — so a stock install draws labels without "unknown font family" warnings. Override it like any board default:

```
#set-board-defaults(label-font: "Segoe UI")   // or a list, e.g. ("Helvetica",
"DejaVu Sans Mono")
```

3 Diagrams

`chess-diagram(source, ..)` wraps a board in a `#figure` (kind `"chess"`), so — unlike a bare `board` — it is captioned, counted, referenceable, and listed by an outline. `source` is the same FEN / position / squares the board takes, and it accepts every board option from the previous chapter.

A diagram carries two optional labels: a **game-info** line above the board (drawn automatically as `"<White> – <Black> (<Year>)"` when both players are known) and the figure **caption** below it.

```
#chess-diagram(
  "r1bqkbnr/pppp1ppp/2n5/4p3/4P3/5N2/
PPPP1PPP/RNBQKB1R",
  white: "Morphy", black: "NN", year:
1858,
  caption: [After 2...Nc6],
  size: 4.2cm,
)
```

Morphy – NN (1858)

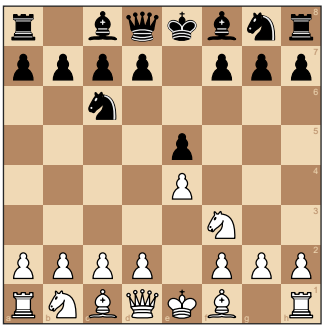


Diagram 2: After 2...Nc6

`chess-diagram` is the standard-chess sugar over the generic `diagram`; both take the same source and overrides as `board`. Extra named arguments are forwarded to `figure` (e.g. `placement: top`).

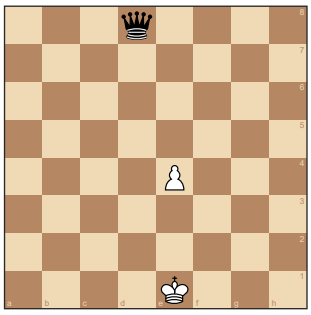
3.1 Boards are not Figures

A bare `board` is plain content: it has **no** caption, **no** figure counter, does **not** resolve `@`-references, and is **not** listed by `chess-diagram-outline`. Only a **diagram** is a figure. So draw a `board` for an inline or decorative position, and a `chess-diagram` whenever you want to caption it, cross-reference it (`@label`), or list it — see **Outlines and references**.

4 Positions

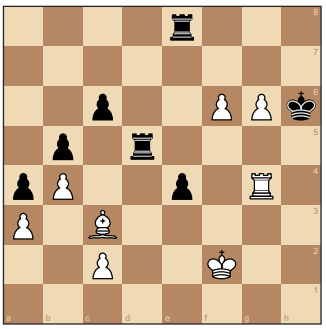
`position(..)` builds the data model — "which piece stands on which square" — that a board or diagram draws. It accepts a FEN string (auto-detected), a **squares dict**, or a **string form**. In a squares dict the piece can be a long name, a kind abbreviation, or a bare letter (UPPER = white, lower = black):

```
#chess-diagram(
  position((
    e1: (kind: "king", color: "white"), // long
    d8: (kind: "q", color: "black"),    // kind
    e4: "P",                            // letter
  )),
  size: 4cm,
)
```



The **string form** reads like the board itself — first line is the TOP rank, `.` is empty. Pass it as a raw block, as below — the most legible, least error-prone way, with no per-line quotes or commas (several row strings work too). It is rectangular-only (this is what lets a board be non-8×8) and rejects characters that aren't a valid piece abbreviation or `.`:

```
#chess-diagram(
  position(```
    ....r...
    ........
    ..p..PPk
    .p.r....
    pP..p.R.
    P.B.....
    ..P..K..
    ........
  ```),
  size: 4.2cm,
)
```



`position` returns a dict `(variant, cols, rows, squares, turn, castling, en-passant, halfmove, fullmove)`; `parse-fen` returns the same shape. The `cols` / `rows` are counted from the string form (otherwise the 8×8 default).

## 5 Games (PGN)

THe predominant form of the distribution and publications of chess games (atleast for western chess) are *PGN files*. PGN stands for **Portable Game Notation** and is a text format for chess games. It is human-readable, and can be parsed by chess software. PGN files contain the moves of a game, along with metadata such as player names, event, date, and result. PGN files cao contain just one or many games.

The `parse-pgn` function returns an **array of games**; `.first()` takes the first one. Read an external file with `read` in your own file, or pass an inline raw block. The examples below assume an already parsed game is in scope:

```
#let game = parse-pgn(read("game.pgn")).first()
```

Parsing is **lazy**: the roster (`tags`), the `result` , and the verbatim `movetext-raw` are extracted cheaply; the move tree is built on demand by `movetext(game)` ; the move parser / generator engine runs only when you ask for a position. So a tournament file read only for results and never tokenises movetext.

`chess-notation(game)` renders the moves (as text) the game already holds, and `diagram-after(game, loc)` renders a **diagram** (a referenceable `#figure` , like `chess-diagram`) of the position at a locator:

```
#chess-notation(game)
```

1.e4 e5 2.Nf3 Nc6 3.Bb5 a6 4.Ba4 Nf6 5.O-O Be7 6.Re1 b5 7.Bb3 d6 8.c3 O-O

```
#diagram-after(game, "3w", size: 4cm)
```

Alice – Bob (2024)

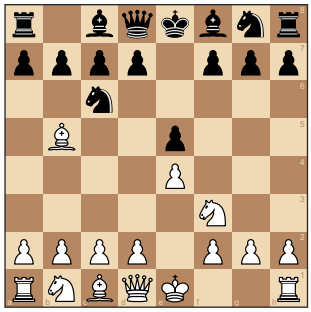


Diagram 5: Position after move 3. Bb5
```

5.1 Locators

A **locator** addresses one position in a game. The simple form is a string — `"30w"` / `"30b"`, the position after White's / Black's 30th **mainline** move. `position-after(game, loc)`, `diagram-after(game, loc, ..)`, and `to-fen(game, locator: ..)` all take this simple string form.

To address a move **inside a variation** (a PGN 'Recursive Annotation Variantion' or RAV), pass a **path** dict instead: `(line: (..hops..), at: "<final move>")`. Each hop is `(at: "<move>", into: <n>)` — branch off at mainline move `at`, descending `into` that move's variation number `n` (**0-based**: `0` is the first variation recorded at that move, `1` the second, …). The top-level `at` is where you stop within the line you reached:

```
#let g = parse-pgn(
  "[White \"V\"][Black \"T\"] 1. e4 (1.
d4 d5 2. c4) e5 *",
).first()
// into variation 0 at White's move 1
(the
// 1.d4 line), position after 2.c4:
#diagram-after(
  g,
  (line: ((at: "1w", into: 0),), at:
"2w"),
  size: 3.4cm,
)
```

V – T

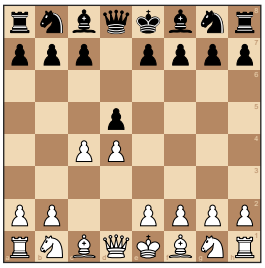


Diagram 6: Position after move 2. c4

Two easy traps:

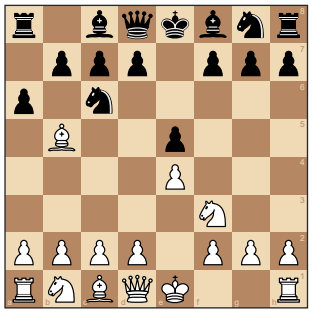
- **The trailing comma.** A single-hop path is written `((at: "1w", into: 0),)` — the comma makes it a one-element **array** of hops. Without it, Typst reads a lone parenthesised dict and the locator is malformed.
- **Mainline is the fast path.** The string form (equivalently an empty `line: ()`) indexes a memoised list of mainline positions, so many diagrams off one game stay cheap; the path form is walked move by move.

Addressing a variation that isn't there (`into:` past the recorded count) or a move past the end of its line is a hard error.

5.2 Playing Moves onto a Position

To explore a **new** line, or build a position from a FEN plus some moves, use `chess-moves(source, moves)`. `source` is `none` (the standard start), a FEN string, or a position; `moves` is move text or a SAN array. It resolves each move against the legal moves (illegal/ambiguous is a hard error) and returns the **final** position, never mutating the source:

```
#chess-diagram(
  chess-moves(none, "1. e4 e5 2. Nf3
Nc6 3. Bb5 a6"),
  size: 4cm,
)
```



5.3 Notation Output

For chess publications, notational output of the move text is as important as showing board positions. This output has to be flexible and localisable. While you sometimes want to show move text exactly as it was

recorded in the PGN, you often want to amend and reformat the move text for your own purposes. The `notation(..)` function is the workhorse for this. It takes a game, a move-text string, or a SAN array and produces a formatted move text output. It can localise the piece letters and render figurine glyphs, and it can include or exclude move numbers, results, NAGs, comments, and embedded diagrams.

To support different chess variants in the future `notation(..)` is the **variant-agnostic** primitive and `chess-notation(..)` is the **standard-western-chess** wrapper over it — identical output today, but `chess-notation` fixes the variant and rejects a source of a non-standard `variant`. The split is deliberate and forward-looking: staunton is **variant-forward**, so a future variant gets its own name — a planned `xiangqi-notation`, `shogi-notation`, … — each a thin wrapper over the same generic `notation` core, while `chess-notation` stays western-chess-specific. The same pairing runs through the package's drawing and notation entry points — `board` / `chess-board`, `diagram` / `chess-diagram`, `notation` / `chess-notation`: reach for the `chess-` name for ordinary chess, and the generic one only when you are deliberately variant-agnostic. Two functions stand slightly apart: `position` is variant-parameterised (the variant rides on its source or `variant:` argument, so there is no separate `chess-position`), and `chess-moves` is chess-only today (the engine does standard chess, so there is no generic `moves`). Both this manual and everyday use favour `chess-notation`.

Either formats move text the game already holds — no engine. Its `source` is a game, a move-text string, or a SAN array; it localises the piece letters and renders figurine glyphs:

```
#chess-notation(game, lang: "de")
```

1.e4 e5 2.Sf3 Sc6 3.Lb5 a6 4.La4 Sf6 5.O-O Le7 6.Te1 b5 7.Lb3 d6 8.c3 O-O

```
#chess-notation(game, figurine: true)
```

1.e4 e5 2.♘f3 ♞c6 3.♗b5 a6 4.♗a4 ♞f6 5.O-O ♝e7 6.♖e1 b5 7.♗b3 d6 8.c3 O-O

`from` / `to` restrict output to an **inclusive** slice of moves. Both are the simple `"8b"` / `"12w"` locators; `from` defaults to the first move, `to` to the last:

```
#chess-notation(game, from: "2w", to: "3b")
```

2.Nf3 Nc6 3.Bb5 a6

`from` / `to` bound a slice of the **mainline**: they are the simple `"8b"` / `"12w"` locators, not the variation **path** form that `diagram-after` takes (rendering variations is a separate control — see the next section). A `from` past the end or a `to` before `from` is a hard error.

Other options: `move-numbers`, `result`, `bold-mainline` (render the mainline moves bold to set them off from variations — see **Variations**), and — for a **game** source — `nags` / `comments`. The last three consult the PGN-handling defaults. Localization substitutes only the piece letters; files, ranks, captures, check marks and `O-O` are untouched.

5.3.1 Variations

By default `notation` renders only the **mainline**. Set `variations: true` (or the `set-pgn-defaults(variations: ..)` document default) to splice the game's variations (RAVs) into the output — in parentheses, correctly numbered: a white-first line reads `3.Bc4 …`, a black-first line `3...Bc5 …`, and the resumed mainline move re-shows its number:

```
#chess-notation(
  parse-pgn(
    "1. e4 e5 2. Nf3 Nc6 3. Bb5 (3. Bc4 Bc5) a6 *",
  ).first(),
  variations: true,
)
```

1.e4 e5 2.Nf3 Nc6 3.Bb5 (3.Bc4 Bc5) **3...a6**

Variations nest to any depth and honour `nags` / `comments` / `figurine` / `lang` inside the parentheses. `variation-style: "block"` keeps the parentheses but breaks each variation onto its own line, indented one level per nesting depth (a variation that ends on a nested line closes with a `)` on its own line) — the analysis-view layout:

```
#chess-notation(
  parse-pgn(
    "1. e4 (1. d4 d5 (1... Nf6 2. c4)) e5 *",
  ).first(),
  variations: true,
  variation-style: "block",
)
```

1.e4
(1.d4 d5
(1...Nf6 2.c4)
)
1...e5

To render **one specific** variation on its own, pass `line:` — a path locator (the same `diagram-after` shape, or just its hops array) that descends into the variation. It is numbered from its real branch ply, so it reads exactly as you'd write it in analysis:

```
#let g = parse-pgn(
  "1. e4 e5 2. Nf3 Nc6 (2... d6 3. d4) 3. Bb5 *",
).first()
// the 2...d6 side line, on its own:
#chess-notation(g, line: ((at: "2b", into: 0),))
```

2...d6 3.d4

Each hop is `(at: "<move>", into: <n>)` — nest hops to reach a deeper line. The addressed line's **own** variations follow the `variations` flag. (`from` / `to` stay mainline-only and don't combine with `line`.)

5.3.2 Building on a Game: NAGs and Comments

Beyond `nags: true` (which renders the `$n` NAGs a game **already** carries), you can attach NAGs and comments **programmatically** — without editing the PGN — and then render them like any parsed game. `with-nags(game, ..)` and `with-comments(game, ..)` each return a **new** game (the source is never mutated), so they compose:

```
#let g = parse-pgn(
  "1. e4 e5 2. Nf3 Nc6 3. Bb5 (3. Bc4 Bc5) a6 *",
).first()
#chess-notation(
  with-comments(
    with-nags(g, ("3w": "!")),
    (((line: ((at: "3w", into: 0),), at: "3w"), "a sharp try"),),
  ),
  variations: true, nags: true, comments: true,
)
```

1.e4 e5 2.Nf3 Nc6 3.Bb5! (3.Bc4 a sharp try Bc5) **3...a6**

Moves are addressed the same way everywhere — a **mainline** locator (`"3w"`) or a variation **path** dict — so you can annotate a move **inside** a (nested) variation, not only the mainline. Two input forms:

- a **dict** `("3w": "!", "5b": "$14")` — mainline moves only (dict keys are strings);
- an **array of** `(locator, value)` **pairs** — any move, including a path-dict locator (as in the example above).

`with-nags` values are a `"$n"` code or a suffix glyph (`! ? !! ?? !? ?!` , sugar for `$1` – `$6`), or an array of those; `with-comments` values are plain-text strings. Both **replace** what was on the move. Comments show only with `comments: true` , NAGs only with `nags: true` .

5.3.3 Adding Variations

`with-variation(game, at:, moves:)` grows the tree: it adds a variation as an **alternative** to the move at `at` (a mainline locator or a path dict). `moves` is a PGN movetext fragment — the same syntax `parse-pgn` reads — so one call can carry move numbers (recomputed), nested `()` variations, `$n` NAGs, and `{comments}` ; a plain SAN run like `"Bc4 Bc5"` is the simplest case:

```
#let g = parse-pgn("1. e4 e5 2. Nf3 Nc6 3. Bb5 a6 *").first()
#chess-notation(
  with-variation(g, at: "3w",
    moves: "3. Bc4 Bc5! (3... Nf6 4. d4) {a sharp alternative}"),
  variations: true, nags: true, comments: true,
)
```

1.e4 e5 2.Nf3 Nc6 3.Bb5 (3.Bc4 Bc5! a sharp alternative (3...Nf6 4.d4)) **3...a6**

The variation is **appended** to the move's variations (its index `into` is the previous count — `0` for a move with none yet), so you can address into it afterwards and it composes with `with-nags` / `with-comments` . Together these let you build a whole annotated tree from a bare game, then render or navigate it exactly like a parsed PGN. Moves are **not** checked for legality when added — an illegal move surfaces only if you navigate into the line (`diagram-after`), matching the rest of the lazy model.

5.4 Exporting FEN

`to-fen` is the inverse of `parse-fen` . It serialises a position, or a game at a locator. Standard 8×8 positions round-trip exactly:

```
#raw(to-fen(chess-moves(none, "1. e4 e5 2. Nf3")))
```

```
rnbqkbnr/pppp1ppp/8/4p3/4P3/5N2/PPPP1PPP/RNBQKB1R b KQkq - 1 2
```

5.5 Drawing Annotations in PGNs

PGN comments can carry drawing annotations — `[%cal …]` for arrows, `[%csl …]` for highlights. Processing is **off by default**; opt in per call with `annotations: true` (or document-wide with `set-pgn-defaults` — see Section 5.6). The demo game annotates its 2nd move:

```
// move 2: {[%cal Gf3e5] [%csl Re5]}
#diagram-after(game, "2w", annotations:
true, size: 4cm)
```

Alice – Bob (2024)

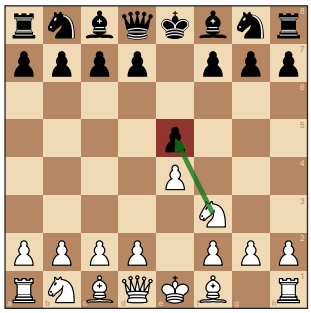


Diagram 8: Position after move 2. Nf3

The color letters (`G R Y B O`) resolve through the `annotation-colors` board style; annotations merge with any `arrows` / `highlight` you pass explicitly.

PGN marks and your own marks, combined. The two sources add up — the marks a PGN comment carries (`annotations: true`) and the `arrows` / `highlight` you pass on the same call are drawn **together** on one board. So you can take an author's annotated game and layer your own emphasis on top without editing the PGN. Here the green `f3→e5` arrow and red `e5` highlight come from the comment, while the `b1→c3` arrow and the circle on `d4` are added programmatically:

```
#diagram-after(game, "2w",
  annotations:
true,                                  //
Gf3e5 + Re5, from the PGN
  arrows: (("b1",
"c3"),),                             //
added here
  highlight: ((square: "d4", shape:
"circle"),),  // added here
  size: 4cm,
)
```

Alice – Bob (2024)

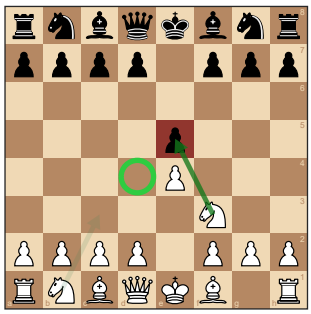


Diagram 9: Position after move 2. Nf3

With embedded diagrams. The `diagrams` switch (PGN handling) makes `notation(diagrams: true)` splice a board after each move whose comment carries a diagram marker. If that **same** comment also holds `%cal` / `%csl` **and** `annotations` is on, the spliced board shows those arrows/highlights too — the two switches compose:

```
// 2. Nf3 {[%cal Gf1c4] [%csl Re5] #[After 2.Nf3]} Nc6 ...
#set-pgn-defaults(diagrams: true, annotations: true)
#chess-notation(game)   // ...text, then a board after 2.Nf3 with the arrow +
highlight
```

`diagrams` decides **whether** a board appears; `annotations` decides whether it carries the marks. The marker and the `%cal` / `%csl` must be in the **same** move's comment. (Only mainline moves are addressed — `notation` renders the mainline.)

5.6 PGN Handling

The switches above — `annotations`, `nags`, `comments`, `diagrams`, `variations`, `bold-mainline` — form the **PGN-handling** group: they decide how much of a parsed game's embedded extras get interpreted at render time. Parsing itself stays lossless; these only decide what is **processed**, and (except `bold-mainline`) **all default off**. Each is a per-call argument (`auto` → the document default) on `notation` / `diagram-after`, or a document-wide default via `set-pgn-defaults`:

```
#set-pgn-defaults(annotations: true, nags: true, comments: true)
```

key	effect
`annotations`	`%cal` / `%csl` → arrows/highlights on `diagram-after`
`nags`	render NAGs (`Nf3!`, `d4±`) in `notation`
`comments`	include comment prose in `notation`
`diagrams`	embed a board in `notation` after each move marked for one
`variations`	splice variations (RAVs) into `notation`, in parentheses
`bold-mainline`	render `notation` mainline moves bold (variations stay normal)

Why most of these default off. For a typical publication you want clean, readable move text: the mainline, set as prose. A PGN's **embedded diagrams**, **drawing annotations**, and free-form **comments** are usually working notes that would clutter that output, so they stay off unless you deliberately ask for them. **NAGs** sit in between — a `!` or `±` is often wanted in a published line — but they, too, default off so nothing appears that you didn't request. **Variations**, by contrast, are a first-class part of chess writing: staunton fully supports them, and whether a render shows **only the mainline** or **the variations as well** is exactly the choice `variations` gives you (per call, or document-wide) — see **Variations** under Section 5. In short: parsing keeps everything; rendering shows only what you opt into.

`set-chess-defaults` routes these same keys through its umbrella, alongside the board, diagram, table and language buckets — see Section 8.

5.7 Errors

Malformed PGN is a **hard error**: broken tag syntax and stray variation parens fail at parse time; illegal, ambiguous, or unparseable moves fail when the position is navigated. Missing Seven-Tag-Roster tags are tolerated (they default).

6 Tournament Tables

Tournament tables are built from a parsed PGN's roster + results (no engine). The `*-table` renderers produce a captioned, referenceable `#figure` (kind `"chess-table"`). Each works on a list of games, with `by: "player"` or `by: "team"`. The examples use a parsed 4-player round-robin in `games`:

```
#let games = parse-pgn(read("event.pgn"))
```

`standings-table` sorts best-first (score, then tie-breaks, then first appearance):

```
#standings-table(games, by: "player")
```

Pos	Player	Pl	+	=	−	Pts	Bh	SB
1	Alice	3	2	1	0	2½	3½	2½
2	Dave	3	1	2	0	2	4	2½
3	Bob	3	1	0	2	1	5	½
4	Carol	3	0	1	2	½	5½	1

`crosstable-table` renders the round-robin grid (it **requires** a complete round-robin and errors otherwise — use standings + progress for Swiss/league):

```
#crosstable-table(games, by: "player")
```

#	Player	1	2	3	4	Pts
1	1 Alice		½	1	1	**2½**
2	2 Dave	½		1	½	**2**
3	3 Bob	0	0		1	**1**
4	4 Carol	0	½	0		**½**

`progress-table` shows the round-by-round running score (needs the `Round` tag):

```
#progress-table(games, by: "player")
```

#	Player	R1	R2	R3	Pts
1	Alice	1 (1)	1 (2)	½ (2½)	**2½**
2	Dave	½ (½)	1 (1½)	½ (2)	**2**
3	Bob	0 (0)	0 (0)	1 (1)	**1**
4	Carol	½ (½)	0 (½)	0 (½)	**½**

Each renderer takes `caption` (used by refs and the outline), `title` (a heading above the table), `supplement`, and `lang`. The compute functions (`standings`, `crosstable`, `progress`) are also public, returning plain data for custom layouts. **Team** mode groups games by the `Round = "round.board"` convention into matches.

7 Outlines and References

Diagrams and tables each carry a distinct figure `kind` (`"chess"` / `"chess-table"`), so they get their own counters, can be **referenced** (`@label` → "Diagram 3" / "Table 2"), and can be **listed separately**:

```
#chess-diagram-outline()      // list of chess diagrams
#chess-table-outline()        // list of tournament tables
#chess-outlines()             // both, diagrams then tables

#chess-diagram(starting-fen, caption: [Start]) <start>
As shown in @start, ...
```

Only figures that carry a **caption** appear in an outline (a caption-less figure is still referenceable but unlisted, matching Typst). Titles are language-aware by default and settable per call (`title:`, `lang:`) or document-

wide (`set-diagram-defaults(outline-title: ..)` — see Section 8). Remember that a bare `board` is not a figure, so it can be neither referenced nor listed — use a `chess-diagram`.

8 Document-Wide Defaults

Package **staunton** features sensible default settings out of the box, you rarely need to adjust them at all. But if you need to, you can achieve that in two ways: per-call arguments overriding the default settings, and **document-wide** defaults that can be set with `set-chess-defaults` or the more specific setters. Adjusting settings this way affects **every subsequent** diagram and table unless overridden per-call.

Defaults live in **five** buckets, each with its own setter:

bucket	setter	controls
board	`set-board-defaults`	square colors, labels, piece set, highlights, arrows, grid, size — full list in Section 10.2
diagram	`set-diagram-defaults`	the diagram figure: game-info line, supplement, outline title
table	`set-table-defaults`	the table figure: supplement, outline title, title gap
language	`set-lang`	the document language (localized strings)
PGN handling	`set-pgn-defaults`	what a parsed game's extras render — see Section 5.6

`set-chess-defaults` is the single **umbrella** over **all five** buckets: pass any key from any bucket — **including** `lang` — and it is routed to the bucket that owns it.

```
#set-board-defaults(light: rgb("#eeeed2"), dark: rgb("#769656"), size: 5cm)
#set-lang("de")
#set-pgn-defaults(nags: true)
// ...or all of the above at once, through the umbrella:
#set-chess-defaults(dark: blue, lang: "de", nags: true, info-gap: 1em)
#set-piece-set("merida")    // sugar for set-board-defaults(piece-set: ..)
```

Every default is equivalently a per-call argument — the same green theme, set once above vs. passed to one diagram:

```
#chess-diagram(
  "rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/
RNBQKBNR",
  light: rgb("#eeeed2"), dark:
rgb("#769656"),
  size: 4cm,
)
```

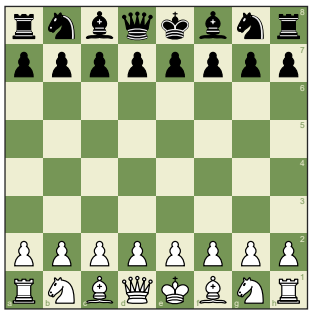


Diagram 10: Position at move 1, White to play

Two caveats. `flip` is **not** allowed in any defaults setter — orientation is a per-board choice, so `set-chess-defaults(flip: ..)` is an error. And `supplement` / `outline-title` live in **both** the diagram and table buckets; the umbrella routes them to **diagram**, so use `set-table-defaults` for the table ones.

8.1 Language

Package **staunton** supports localisation of text-related output. At the moment we support six different languages; apart from the standard English, we offer German, French, Spanish, Italian, Portuguese, and Russian. We can easily extend the list of supported languages by adding new translation files.

The `notation` / `chess-notation` functions localise the piece letters, and the `chess-diagram` / `chess-table` figures carry language-aware titles and captions. The `lang:` argument on each function overrides the document default, and the document default is set with `set-lang`.

A single document **language** drives every language-aware string — diagram and table supplements, outline titles, and notation piece letters. Default is English; `"auto"` follows `#set text(lang: ..)`; or pick a code:

```
#set-lang("de")      // Diagramm / Tabelle / Sf3, Lb5, ...
#set-lang("auto")    // follow #set text(lang: ..)
```

Every localizable string is also per-call overridable (the `lang:` argument seen on `chess-notation` above). Adding a language is a no-code change: drop a `src/assets/i18n/<code>.typ` and register it in `src/i18n.typ`.

8.2 PGN Handling

The fifth bucket, **PGN handling**, decides how much of a parsed game's embedded extras (NAGs, comments, annotations, variations, embedded diagrams) get interpreted at render time. Because those switches are best understood next to the notation and board features they govern, they are documented with the games themselves — see Section 5.6. `set-pgn-defaults` sets them document-wide, and `set-chess-defaults` routes the same keys through the umbrella.

9 HTML Export

Besides paged output (PDF / PNG), **staunton** also supports Typst's **HTML export**. Compile with the `html` feature and target:

```
typst compile --features html --format html your-doc.typ out.html
```

The library-level output maps onto native HTML:

- **move notation** becomes ordinary inline text (mainline moves as `<strong>`);
- **tournament tables** become real `<table>` elements;
- **diagram** and **table** figures keep their captions, numbering and supplements as `<figure>` / `<figcaption>`, and `@`-references and the diagram / table **outlines** resolve to in-document links;
- **boards and diagrams** are embedded as **inline SVG** (a board is layout-drawn, so it is wrapped in `html.frame` under an HTML target — paged export is unchanged).

Caveats. Typst's HTML export is itself *under active development*, so this is **limited** support, not full PDF parity. Page-level chrome (the `#set page` header / footer, `#pagebreak`, multi-column `#grid` layout) is dropped by HTML export — that is document styling, not part of staunton. Treat HTML output as a useful secondary format that will improve as Typst's own HTML support matures.

10 Common Parameters

This chapter collects the recurring **argument value shapes** and the full board / diagram / table option lists — the parameters that many functions share. The **Main functions** and **Advanced functions** chapters that follow then give every public function's signature with each parameter's type and default, generated directly from the source docstrings. The guide chapters above show each feature in use with a rendered example; this reference part answers "what exactly can I pass".

10.1 Argument Value Shapes

A few arguments accept more than one shape and recur across functions, so they are described once here.

`source` for `board`, `chess-board`, `diagram`, `chess-diagram`, `position` — a **FEN string** `"rnbqkbnr/…"`; a **position** dict (from `position` / `parse-fen`); a **squares** dict `(e1: "K", d8: (kind: "q", color: "black"), e4: "P")` (piece as a long name, kind abbreviation, or bare letter — UPPER white, lower black); or the **string form** (rank-per-line rows, `.` = empty; one raw block or several row strings). `chess-*` reject a non-standard `variant`.

`locator` for `position-after`, `diagram-after`, `to-fen`, `move-san`, `move-node`, and builder addresses — a **mainline** string `"12w"` / `"12b"`, or a **path** dict `(line: (..hops..), at: "<move>")`, each hop `(at: "<move>", into: <n>)` (descend into variation `n` at that move), to reach a move inside a (possibly nested) variation. `notation`'s `line:` also takes a bare hops array; its `from` / `to` are mainline strings only.

`size` a `length` (default: `4cm`), a `ratio` of the available width (default: `60%`), or `auto`.

`highlight` entry a square name `"e4"`; a `(square, color)` pair; or a dict `(square:, shape:, color:)` with shape `"filled"` / `"cross"` / `"circle"`.

`arrows` entry `(from, to)`; `(from, to, color)`; or a dict `(from:, to:, color:)`.

builder overrides for `with-nags` / `with-comments` — a **dict** of mainline locators (`("3w": v)`) or an **array of `(locator, value)` pairs** (this form also takes path-dict locators). A NAG value is `"$n"` or a suffix glyph `! ? !! ?? !? ?!` (sugar for `$1` – `$6`), or an array of those; a comment value is a plain string.

annotation color letters for PGN `%cal` / `%csl` and the `annotation-colors` map — `G` `R` `Y` `B` `O`.

10.2 Board Style Options

Accepted by `board` / `chess-board` / `diagram` / `chess-diagram` per call, and by `set-board-defaults` / `set-chess-defaults` document-wide (see Section 8).

option	default	meaning
`size`	`auto`	board size: a `length`, a `ratio` of the width, or `auto`
`light` / `dark`	tan theme	the two square fill colors
`labels`	`true`	show rank/file labels
`label-font`	`("Arial", "DejaVu Sans Mono")`	label font — a family or a fallback list
`label-mode`	`"on-square"`	`"on-square"` / `"outside"` / `"border"`
`file-side` / `rank-side`	`bottom` / `right`	which edge files / ranks sit on
`file-label-corner` / `rank-label-corner`	`left` / `right`	on-square label corner

option	default	meaning
`border-theme`	`"square"`	`"border"` band theme: `"square"` / `"brown"` / `"dark"`
`border`	`0.5pt + luma(40)`	thin board outline (`none` to drop)
`grid`	`false`	1pt grid lines between squares
`piece-set`	`"cburnett"`	SVG set name, or `"unicode"` for the glyph fallback
`piece-scale`	`0.95`	fraction of a square a piece occupies
`highlight` / `arrows`	`()`	squares / arrows to draw — see the value shapes
`highlight-shape`	`"filled"`	default shape for plain-string highlight entries
`highlight-fill` / `highlight-transparency`	green, `75%`	filled-highlight color and its transparency
`cross-color` / `circle-color`	red / green	cross / circle stroke colors
`cross-width` / `circle-width`	`2pt`	cross / circle stroke widths
`arrow-color` / `arrow-transparency`	green, `85%`	default arrow color and its transparency
`arrow-width`	`auto`	arrow shaft width; `auto` scales with the square
`annotation-colors`	G/R/Y/B/O map	PGN `%cal` / `%csl` color-letter ⟶ color
`label-color`	`luma(90)`	`"outside"` -mode strip label color
`label-border-ratio`	`0.07`	`"border"` -mode band width, as a board fraction
`white-fill` / `black-fill` / `piece-font`	—	the `"unicode"` glyph fallback only
`baseline-inset`	`0.20`	glyph fallback: baseline lift (square fraction)

10.3 Diagram, Table, Language, and PGN-Handling Options

Diagram (`set-diagram-defaults`): `info-bold` (`true`), `info-gap` (`0.6em`), `supplement` (`auto` ⟶ localized “Diagram”), `outline-title` (`auto`).

Table (`set-table-defaults`): `supplement` (`auto` ⟶ “Table”), `outline-title` (`auto`), `title-gap` (`0.6em`).

Language (`set-lang`): `lang` — `"en"` (default), a code (`"de"` , `"es"` , `"fr"` , `"it"` , `"pt"` , `"ru"`), or `"auto"` (follow `#set text(lang: ..)`).

PGN handling (`set-pgn-defaults`): `annotations` , `nags` , `comments` , `diagrams` , `variations` — all `false` by default; `bold-mainline` defaults `true` (see PGN handling).

11 Main Functions

The everyday API, grouped by task. Each entry is generated from the function's source docstring: its signature, then every parameter with its type and default.

11.1 Boards and Diagrams

board

Draw a bare board — no caption, no figure — the variant-agnostic drawing primitive that every diagram builds on. The variant (if any) rides on `source`; the variant-named wrappers (`chess-board`, a future `xiangqi-board`, …) are thin sugar over this.

Parameters

```
board(
  source: str | dictionary,
  flip: bool,
  ..overrides: arguments
) -> content
```

source `str` or `dictionary`

the position to draw — a **FEN string**, a **position** dict (from `position` / `parse-fen`), or a bare **squares** dict (`(e1: "K", …)`).

flip `bool`

show the board from Black's side. Per-call only — never a document default.

Default: `false`

..overrides `arguments`

any board **style** option (`size`, `light`, `dark`, `labels`, `label-mode`, `file-side`, `rank-side`, `piece-set`, `highlight`, `arrows`, `grid`, …) — see Board style options.

chess-board

Standard western-chess board — the variant-named sugar over `board`, and the everyday entry point. Identical rendering, but it documents the variant and rejects a non-standard `source`. (Other variants get their own entry, e.g. a future `xiangqi-board`.)

Parameters

```
chess-board(
  source: str | dictionary,
  flip: bool,
  ..overrides: arguments
) -> content
```

source `str` or `dictionary`

a standard-chess position — a FEN string, a position dict, or a squares dict.

flip `bool`

show the board from Black's side.

Default: `false`

..overrides `arguments`

any board **style** option (as for `board`).

diagram

A board wrapped in a `#figure` — the variant-agnostic diagram, and the generic primitive under `chess-diagram` (and a future `xiangqi-diagram`). Draws an automatic "White – Black (Year)" line above when both players are known, and a default caption below for a FEN source ("Position at move N, X to play").

Parameters

```
diagram(
  source: str | dictionary,
  white: str | none,
  black: str | none,
  event: str | none,
  year: int | str | none,
  caption: auto | content | none,
  game-info: auto | content | none,
  flip: bool,
  lang: auto | str,
  ..args: arguments
) -> content
```

source `str` or `dictionary`

a FEN string, a position dict, or a bare board dict.

white `str` or `none`

white player's name, for the automatic info line.

Default: none

black `str` or `none`

black player's name.

Default: none

event `str` or `none`

event name (carried; not shown by default).

Default: none

year `int` or `str` or `none`

year, appended to the info line in parentheses.

Default: none

caption `auto` or `content` or `none`

the figure caption; `auto` is the source-specific default (a FEN gets "Position at move N…"; a position or board dict gets none).

Default: `auto`

game-info `auto` or `content` or `none`

the above-board line; `auto` is the automatic player line — pass your own content, or `none` to drop it.

Default: `auto`

flip `bool`

show the board from Black's side (per-diagram only).

Default: `false`

lang `auto` or `str`

language for the supplement; `auto` follows the document.

Default: `auto`

..args `arguments`

board **style** options go to the renderer; anything else is forwarded to `#figure` (e.g. `placement: top`).

chess-diagram

Standard western-chess diagram — the variant-named sugar over `diagram` and the everyday entry point. Same behaviour, but it documents the variant and rejects a non-standard `source`.

Parameters

```
chess-diagram(
  source: str dictionary,
  ..args: arguments
) -> content
```

source `str` or `dictionary`

a standard-chess position (FEN, position dict, or squares dict).

..args `arguments`

everything `diagram` accepts (players, `caption`, `game-info`, `flip`, `lang`, style options, and `#figure` arguments).

11.2 Positions

position

Build a `position` — the data a board draws — from manual placement. (A single string containing `/` is treated as a FEN and delegated to `parse-fen`.) Positional arguments place the pieces; named arguments set the metadata.

Pieces are given as a **squares dict** — `(e1: "K", d8: (kind: "q", color: "black"))`, a piece being a bare letter (UPPER white, lower black) or a `(kind, color)` with a long or abbreviated kind — or in the **string form**, one rank per line (or per positional string), `.` for an empty square.

Parameters

```
position(..args: arguments) -> dictionary
```

..args `arguments`

the piece placement (positional), plus named metadata — `variant` (default `"standard"`), `turn`, `castling`, `en-passant`, `halfmove`, `fullmove`, and `cols` / `rows` (default: from the variant, or counted in the string form).

parse-fen

Parse a FEN string into a `position` dict — the inverse of `to-fen`. The result carries the board (square name ⟶ `(kind, color)`) plus the side to move, castling rights, en-passant target, and the halfmove / fullmove clocks.

Parameters

```
parse-fen(fen: str) -> dictionary
```

fen `str`

a FEN record, e.g. `"rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR w KQkq - 0 1"`.

to-fen

Export a position as a FEN string — the inverse of `parse-fen`. Serialises either a **position** dict directly, or a **game** at a locator. Standard 8×8 positions round-trip exactly.

Parameters

```
to-fen(
  source: dictionary,
  locator: str dictionary
) -> str
```

source `dictionary`

a **position** dict, or a **game** (from `parse-pgn`).

locator `str` or `dictionary`

required when `source` is a game — which move's position to serialise (a mainline `"12w"` or a path dict). Ignored for a position.

Default: none

starting-fen

The standard chess starting position, as a FEN string constant (handy as the `source` for `chess-moves`, `parse-fen`, `board`, …).

11.3 Games (PGN)

parse-pgn

Parse PGN text into an array of games. The roster (tags), result, and the verbatim movetext substring are extracted eagerly; the move tree is parsed lazily on first use via `movetext(game)`, so a document that shows only a few positions never parses the rest.

Parameters

```
parse-pgn(input: str | content) -> array
```

input `str` or `content`

the PGN source — a string, or a raw block (```…```) / `#raw(..)` .

movetext

The parsed movetext tree (an array of move nodes) of a game — the mainline spine with recursive `variations`. Built on demand from the game's raw movetext and memoised, so deeper structure errors (e.g. a `(` without a preceding move) surface here.

Parameters

```
movetext(game: dictionary) -> array
```

game `dictionary`

a parsed game (from `parse-pgn`).

mainline

The mainline of a game as an array of SAN strings (the game as played).

Parameters

```
mainline(game: dictionary) -> array
```

game `dictionary`

a parsed game (from `parse-pgn`).

game-result

The game result token: `"1-0"`, `"0-1"`, `"1/2-1/2"`, or `"*"`.

Parameters

```
game-result(game: dictionary) -> str
```

game `dictionary`

a parsed game (from `parse-pgn`).

position-after

The position at a locator — handles the mainline and (nested) variations.

Parameters

```
position-after(
  game: dictionary,
  locator: str | dictionary
) -> dictionary
```

game `dictionary`

a parsed game (from `parse-pgn`).

locator `str` or `dictionary`

a mainline `"30w"` / `"30b"`, or a variation path dict `(line: (..hops..), at: "<move>")`.

diagram-after

A chess diagram for the position at `locator` within a parsed game. Players and year default to the game's roster tags (so the info line is automatic) and the caption defaults to "Position after move …" (the move played).

When the resolved PGN-handling `annotations` switch is on, `%cal` / `%csl` drawing annotations in the move's comment become arrows / highlights, merged with any passed explicitly.

Parameters

```
diagram-after(
  game: dictionary,
  locator: str dictionary,
  white: auto str none,
  black: auto str none,
  year: auto int str none,
  caption: auto content none,
  annotations: auto bool,
  flip: bool,
  game-info: auto content none,
  lang: auto str,
  ..args: arguments
) -> content
```

game `dictionary`

a parsed game (from `parse-pgn`).

locator `str` or `dictionary`

which position — a mainline `"30w"` / `"30b"` or a variation path dict.

white `auto` or `str` or `none`

white player; `auto` uses the game's `White` tag.

Default: `auto`

black `auto` or `str` or `none`

black player; `auto` uses the `Black` tag.

Default: `auto`

year `auto` or `int` or `str` or `none`

year; `auto` uses the `Date` tag.

Default: `auto`

caption `auto` or `content` or `none`

`auto` is “Position after move …”.

Default: `auto`

annotations `auto` or `bool`

process `%cal` / `%csl` into arrows / highlights; `auto` consults `set-pgn-defaults` (off by default).

Default: `auto`

flip `bool`

show the board from Black’s side.

Default: `false`

game-info `auto` or `content` or `none`

the above-board line (as for `diagram`).

Default: `auto`

lang `auto` or `str`

language for the supplement; `auto` follows the document.

Default: `auto`

..args `arguments`

board **style** options and `#figure` arguments.

move-san

The SAN of the move addressed by `locator` (e.g. `"O-O-O"`). Used to build PGN-diagram captions.

Parameters

```
move-san(
  game: dictionary,
  locator: str dictionary
) -> str
```

game `dictionary`

a parsed game (from `parse-pgn`).

locator `str` or `dictionary`

a mainline `"30w"` / `"30b"`, or a variation path dict.

move-node

The full move node addressed by `locator` — its SAN plus `nags`, comments (`comment-before` / `comment-after`) and `variations`. Used e.g. to recover PGN `%cal` / `%csl` annotations for a diagram.

Parameters

```
move-node(
  game: dictionary,
  locator: str dictionary
) -> dictionary
```

game `dictionary`

a parsed game (from `parse-pgn`).

locator `str` or `dictionary`

a mainline `"30w"` / `"30b"`, or a variation path dict.

chess-moves

Apply a run of moves to a position and return the **final** position (renderable). Each move is resolved against the position's legal moves, so an illegal or ambiguous move is a hard error naming the offending SAN token. The variant is taken from `source`; non-standard variants error in the engine.

Parameters

```
chess-moves(
  source: none str dictionary,
  moves: str content array
) -> dictionary
```

source `none` or `str` or `dictionary`

the starting point — `none` for the standard start, a FEN string, or a position dict.

moves `str` or `content` or `array`

move text (a string or a raw block; move numbers and a trailing result are tolerated and stripped), or an array of SAN tokens.

11.4 Annotate and Build

with-nags

Attach NAGs to addressed moves (mainline or inside variations), returning a **new** game (the source is not mutated). Each value **replaces** that move's NAGs.

Parameters

```
with-nags(
  game: dictionary,
  overrides: dictionary array
) -> dictionary
```

game `dictionary`

a parsed game (from `parse-pgn`).

overrides `dictionary` or `array`

a dict of mainline locators (`("12w": "!")`) or an array of `(locator, value)` pairs (a locator may be a variation path dict). A value is `"$n"`, a suffix glyph (`! ? !! ?? !? ?!`), or an array of those.

with-comments

Attach text comments to addressed moves (mainline or inside variations), returning a **new** game (the source is not mutated). Each comment **replaces** any existing one and is set as the move's `comment-after` (what notation's `comments` switch renders).

Parameters

```
with-comments(
  game: dictionary,
  overrides: dictionary array
) -> dictionary
```

game `dictionary`

a parsed game (from `parse-pgn`).

overrides `dictionary` or `array`

a dict of mainline locators or an array of `(locator, text)` pairs (a locator may be a variation path dict); each `text` is a plain string.

with-variation

Add a variation (RAV) as an **alternative** to the move at `at`, returning a **new** game (the source is never mutated). The variation is appended to that move's variations (its `into` index is the previous count) and numbered from the move's ply at render time. Legality is checked only if you later navigate into the line (e.g. via `diagram-after`).

Parameters

```
with-variation(
  game: dictionary,
  at: str dictionary,
  moves: str content
) -> dictionary
```

game `dictionary`

a parsed game (from `parse-pgn`).

at `str` or `dictionary`

the move to branch at — a **mainline** locator (`"3w"` / `"3b"`), or a **path** dict `(line: (..hops..), at: "<move>")` to reach a move inside a (possibly nested) variation.

Default: none

moves `str` or `content`

a PGN movetext fragment — a plain SAN run like `"Bc4 Bc5"`, or richer text with nested `()`, `$n` NAGs and `{comments}`.

Default: none

11.5 Notation

notation

Render move notation as text, and — when `diagrams` is on and the source is a game — splice a `chess-diagram` into the flow after each move whose comment carries a diagram marker (ChessBase `#` /

`#[caption]`, Scid `[d]` / `[D]`, `\diagram`, `%%diagram`). Embedded diagrams are made in a context, so they are not individually referenceable. The variant-agnostic formatter under `chess-notation`.

Parameters

```
notation(
  source: dictionary str array,
  ..args: arguments
) -> content
```

source `dictionary` or `str` or `array`

a parsed **game**, a **move-text string**, or a **SAN array**.

..args `arguments`

the formatting options — `from` / `to` (mainline slice), `line` (render one variation), `figurine`, `lang`, `nags`, `comments`, `variations`, `variation-style` (`"inline"` / `"block"`), `bold-mainline`, `move-numbers`, `result`, and the embedding switches `diagrams` `annotations`. The `auto`-defaulting ones consult `set-pgn-defaults`.

chess-notation

Standard western-chess notation — the variant-named sugar over `notation` and the everyday entry point. Same behaviour, but it rejects a non-standard `source` variant.

Parameters

```
chess-notation(
  source: dictionary str array,
  ..args: arguments
) -> content
```

source `dictionary` or `str` or `array`

a parsed game, a move-text string, or a SAN array (standard chess).

..args `arguments`

everything `notation` accepts.

11.6 Tournament Tables

standings-table

Render a standings table as a Typst `#table` figure (kind `"chess-table"`). The data options are exactly as for `standings`.

Parameters

```
standings-table(
  games: array,
  by: str,
  tiebreaks: auto array,
  match-points: dictionary,
  title: none content,
  caption: none content,
  supplement: auto content,
  lang: auto str,
  ..table-args: arguments
) -> content
```

games `array`

parsed games (from `parse-pgn`).

by `str`

`"player"` or `"team"`.

Default: `"player"`

tiebreaks `auto` or `array`

as for `standings`.

Default: `auto`

match-points `dictionary`

as for `standings`.

Default: `(win: 2, draw: 1, loss: 0)`

title `none` or `content`

a title shown above the table.

Default: `none`

caption `none` or `content`

the figure caption (needed for the table to appear in `chess-table-outline`).

Default: `none`

supplement `auto` or `content`

the figure supplement; `auto` is language-aware.

Default: `auto`

lang `auto` or `str`

language for defaults; `auto` follows the document.

Default: `auto`

..table-args `arguments`

forwarded to `#table`.

crosstable-table

Render a round-robin cross-table as a Typst `#table` figure. Columns are numbered to match the row order; the diagonal is shaded.

Parameters

```
crosstable-table(
  games: array,
  by: str,
  match-points: dictionary,
  title: none | content,
  caption: none | content,
  supplement: auto | content,
  lang: auto | str,
  ..table-args: arguments
) -> content
```

games `array`

parsed games (from `parse-pgn`).

by `str`

`"player"` or `"team"`.

Default: `"player"`

match-points `dictionary`

team match-point values.

Default: `(win: 2, draw: 1, loss: 0)`

title `none` or `content`

a title shown above the table.

Default: `none`

caption `none` or `content`

the figure caption.

Default: `none`

supplement `auto` or `content`

the figure supplement; `auto` is language-aware.

Default: `auto`

lang `auto` or `str`

language for defaults; `auto` follows the document.

Default: `auto`

..table-args `arguments`

forwarded to `#table`.

progress-table

Render a progress table as a Typst `#table` figure: a column per round showing that round's result and the running total, plus a final total.

Parameters

```
progress-table(
  games: array,
  by: str,
  match-points: dictionary,
  title: none | content,
  caption: none | content,
  supplement: auto | content,
  lang: auto | str,
  ..table-args: arguments
) -> content
```

games `array`

parsed games (from `parse-pgn`).

by `str`

`"player"` or `"team"`.

Default: `"player"`

match-points `dictionary`

team match-point values.

Default: `(win: 2, draw: 1, loss: 0)`

title `none` or `content`

a title shown above the table.

Default: `none`

caption `none` or `content`

the figure caption.

Default: `none`

supplement `auto` or `content`

the figure supplement; `auto` is language-aware.

Default: `auto`

lang `auto` or `str`

language for defaults; `auto` follows the document.

Default: `auto`

..table-args `arguments`

forwarded to `#table`.

games-by-event

Split a games array into a dict keyed by the `Event` tag, preserving insertion order within each event. Games without an `Event` tag group under `""`.

Parameters

```
games-by-event(games: array) -> dictionary
```

games `array`

parsed games (from `parse-pgn`).

11.7 Outlines and References

chess-diagram-outline

An outline listing only chess **diagrams** (figures of kind `"chess"`).

Parameters

```
chess-diagram-outline(
  title: auto | content | none,
  lang: auto | str,
  ..args: arguments
) -> content
```

title `auto` or `content` or `none`

the outline title; `auto` is the language-aware "List of Diagrams" (or the document value from `set-diagram-defaults`).

Default: `auto`

lang `auto` or `str`

language for the default title; `auto` follows the document.

Default: `auto`

..args `arguments`

forwarded to `outline` (e.g. `depth`, `indent`).

chess-table-outline

An outline listing only chess **tables** (standings / cross-table / progress figures of kind `"chess-table"`). Only tables that carry a `caption` appear.

Parameters

```
chess-table-outline(
  title: auto | content | none,
  lang: auto | str,
  ..args: arguments
) -> content
```

title `auto` or `content` or `none`

the outline title; `auto` is the language-aware "List of Tables" (or the document value from `set-table-defaults`).

Default: `auto`

lang `auto` or `str`

language for the default title; `auto` follows the document.

Default: `auto`

..args `arguments`

forwarded to `outline`.

chess-outlines

Print both chess outlines back to back: diagrams first, then tables.

Parameters

```
chess-outlines(
  diagram-title: auto | content | none,
  table-title: auto | content | none,
  lang: auto | str,
  ..args: arguments
) -> content
```

diagram-title `auto` or `content` or `none`

title for the diagram outline; `none` drops it.

Default: `auto`

table-title `auto` or `content` or `none`

title for the table outline; `none` drops it.

Default: `auto`

lang `auto` or `str`

language for the default titles; `auto` follows the document.

Default: `auto`

..args `arguments`

forwarded to both `outline` s.

11.8 Document Defaults

set-chess-defaults

Umbrella setter: route each field to the board, diagram, table, i18n or PGN-handling bucket — handy for setting several defaults at once. `flip` `orientation` are rejected (per-diagram only).

Parameters

```
set-chess-defaults(..fields: arguments) -> content
```

..fields	`arguments`

any named style option from the board, diagram, table, i18n or PGN-handling groups; unknown keys error. (`supplement` `outline-title` , shared by diagram and table, route to the diagram bucket — use `set-table-defaults` for the table ones.)

set-board-defaults

Set default **board** style fields for all subsequent boards and diagrams (document-order state, like a Typst `#set`). `flip` is rejected (per-diagram only).

Parameters

```
set-board-defaults(..fields: arguments) -> content
```

..fields	`arguments`

named board style options (`size` , `light` , `dark` , `labels` , `label-mode` , `piece-set` , ...); unknown keys error.

set-diagram-defaults

Set default **diagram** style fields (the `#figure` wrapper) for subsequent diagrams.

Parameters

```
set-diagram-defaults(..fields: arguments) -> content
```

..fields	`arguments`

named diagram style options (`supplement` , `outline-title` , `info-bold` , `info-gap` , ...); unknown keys error.

set-table-defaults

Set default **table** style fields (the `#figure` wrapper around tournament tables) for subsequent `*-table` renderers.

Parameters

```
set-table-defaults(..fields: arguments) -> content
```

..fields	`arguments`

named table style options — currently `supplement` (default “Table”), `outline-title` , `title-gap` ; unknown keys error.

set-pgn-defaults

Set default PGN-handling fields for subsequent notation / diagram output.

Parameters

```
set-pgn-defaults(..fields: arguments) -> content
```

..fields	`arguments`

named switches — `annotations` , `nags` , `comments` , `diagrams` , `variations` , `bold-mainline` ; unknown keys error.

set-lang

Set the document language for all language-aware strings (supplements, outline titles, notation piece letters).

Parameters

```
set-lang(code: str) -> content
```

code `str`

a language code (`"en"` , `"de"` , …) or `"auto"` (follow `#set text(lang: ..)`).

set-piece-set

Set the default piece set for subsequent diagrams — a convenience wrapper over `set-board-defaults(piece-set: name)` .

Parameters

```
set-piece-set(name: str) -> content
```

name `str`

a bundled piece-set name (`"cburnett"` , `"merida"` , …) or a registered custom set.

12 Advanced Functions

A handful of supported **escape hatches** for programmatic use — not the everyday entry points, but part of the public API and safe to build on. They cover three needs: reading tournament **data** to build your own tables, driving the **engine** directly, and dropping a single piece **glyph** or computing a square in your own layout code.

Everything else in `src/` (the position parser's internals, the SAN encoder, the renderer, the comment interpreter, the localization tables, …) is deliberately **not** re-exported from the package: those names are implementation details that may change between releases. If you truly need one, import it directly from its module — e.g. `#import "@preview/staunton:0.1.0/src/coords.typ": square-name` — with the understanding that it carries no stability promise.

12.1 Tournament data

The `*-table` functions in Tournament Tables render standings, cross-tables and progress grids. These return the underlying **data** instead, so you can lay it out yourself.

standings

Compute standings — an array of records sorted best-first: `(rank, name, score, played, wins, draws, losses, buchholz, sonneborn-berger, [board-points for teams])`.

Parameters

```
standings(
  games: array,
  by: str,
  tiebreaks: auto | array,
  match-points: dictionary
) -> array
```

games `array`

parsed games (from `parse-pgn`).

by `str`

`"player"` or `"team"`.

Default: `"player"`

tiebreaks `auto` or `array`

ordered tiebreak keys, or `auto` for the mode default.

Default: `auto`

match-points `dictionary`

team match-point values, `(win:, draw:, loss:)`.

Default: `(win: 2, draw: 1, loss: 0)`

crosstable

Cross-table for a round-robin: `(names, ranks, totals, matrix)`, rows/cols in standings order, `matrix.at(i).at(j)` entity i's score vs j (player: game points; team: board points), `none` on the diagonal.

Errors if the entity does not form a round-robin (some pair never met) — use `standings` + `progress` for Swiss / league events.

Parameters

```
crosstable(
  games: array,
  by: str,
  match-points: dictionary
) -> dictionary
```

games `array`

parsed games (from `parse-pgn`).

by `str`

`"player"` or `"team"`.

Default: `"player"`

match-points `dictionary`

team match-point values.

Default: `(win: 2, draw: 1, loss: 0)`

progress

Per-entity round-by-round progress: `(names, ranks, rounds, cells)`, where `cells.at(i).at(k)` is `(score, cumulative)` for entity i in `rounds.at(k)` (player: game points that round; team: match points). Needs the `Round` tag; works for open / Swiss events too.

Parameters

```
progress(
  games: array,
  by: str,
  match-points: dictionary
) -> dictionary
```

games `array`

parsed games (from `parse-pgn`).

by `str`

`"player"` or `"team"`.

Default: `"player"`

match-points `dictionary`

team match-point values.

Default: `(win: 2, draw: 1, loss: 0)`

12.2 Engine

Generate and apply moves, or test for check — for puzzles, analysis, or conditional rendering.

legal-moves
All fully-legal moves for the side to move in `position`, as an array of move dicts.

Parameters

```
legal-moves(position: dictionary) -> array
```

position	`dictionary`
the position to generate moves for.	

apply
Apply a move to a position, returning the new position — pure (the input is not mutated). Handles captures, en passant, castling (rook too), promotion, castling-rights and en-passant-target updates, and the move clocks.

Parameters

```
apply(
  position: dictionary,
  move: dictionary
) -> dictionary
```

position	`dictionary`
the position to move from.	

move	`dictionary`
a concrete move dict (e.g. from `san-to-move` or `legal-moves`).	

in-check
Is `color`'s king currently in check in `position`?

Parameters

```
in-check(
  position: dictionary,
  color: str
) -> bool
```

position	`dictionary`
the position to test.	

color	`str`
`"white"` or `"black"`.	

12.3 Pieces and coordinates

`piece-content` renders a single piece glyph for use in running prose; the two coordinate helpers convert between square names and `(col, row)` indices for hand-built overlays.

piece-content
Render a single piece as `size`-tall content (no square background) — the SVG piece if the active set has one, else the glyph fallback.

Parameters

```
piece-content(
  kind: str,
  color: str,
  size: length,
  white-fill: color,
  black-fill: color,
  stroke-ratio: float,
  font: array
) -> content
```

kind `str`

one of `piece-kinds`.

color `str`

`"white"` or `"black"`.

size `length`

the glyph font size.

white-fill `color`

fill for white pieces.

Default: `default-white-fill`

black-fill `color`

fill for black pieces.

Default: `default-black-fill`

stroke-ratio `float`

outline width as a fraction of `size`.

Default: `0.03`

font `array`

the glyph-fallback font family list.

Default: `default-piece-fonts`

parse-square

Parse an algebraic square name into zero-based indices, bounds-checked against the board geometry. One leading file letter (a–z) plus a 1+ digit rank; capitalisation does not matter (`"E4"` = `"e4"`). Returns `(col, row)`.

Parameters

```
parse-square(
  square: str,
  cols: int,
  rows: int
) -> dictionary
```

square `str`

the square name, e.g. `"e4"`.

cols `int`

board width (default `8`).

Default: 8

rows `int`

board height (default `8`).

Default: 8

is-dark-square

Is the square dark? `a1` = `(0, 0)` is a dark square in standard orientation.

Parameters

```
is-dark-square(
  col: int,
  row: int
) -> bool
```

col `int`

zero-based column index.

row `int`

zero-based row index.